

Level 6 Production Project	
Risk Register	
Name: Aayam Regmi	Student I.D.: 77466864
Course: BSc (Hons) Computing	Supervisor's Name: Mr. Saroj Sharma
Title of my Project: Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction	
Severity calculation: (Likelihood) x (Impact) Low: 1-5 Medium: 6-12 High: 13-25	

ID	Risk	Risk Description	Likelihood	Impact	Severity Level	Owner	Mitigation	Status
1	Inaccurate or unstable finger tracking due to occlusion	Frequent occlusion due to fast hand movements while playing the piano	Medium (4)	Medium (3)	(12) Medium	Aayam Regmi	Have datasets with controlled tempos and occlusion levels	Closed
2	Poor model performance under varying lighting conditions	Inconsistency due to light level changes	Medium (3)	Medium (3)	(9) Medium	Aayam Regmi	Conduct all data collection under controlled lighting environment (dim/standard/bright)	Closed
3	Bias or degraded performance across skin tones and hand sizes	Performance disparity in training data categorized by Fitzpatrick scale	Medium (3)	Medium (3)	(9) Medium	Aayam Regmi	Have diversity in participant recruiting in hand sizes and skin tones and perform statistical tests on Fitzpatrick scale groups	Closed
4	Synchronization issues between video & MIDI ground truth	Unsynchronized comparison due to timestamp mismatch	Low (2)	High (5)	(10) Medium	Aayam Regmi	Use precision timestamps and validate test data using annotations	Closed
5	Real-time performance below target (FPS < 15-20)	Low model performance on hardware	Low (2)	Low (2)	(4) Low	Aayam Regmi	Use lighter models and use NVIDIA CUDA acceleration supported models or just multi core CPU procession	Closed
6	Ethical approval or participant consent delays	Delay in approval of research or issues with data	Medium (3)	High (5)	(15) High	Aayam Regmi	Prepare ethical consent form and	Closed

		consent and protection					discuss with supervisor regularly	
7	Hardware failure, malfunction or unavailability	Hardware failure due to various reasons like static charge, water damage or unavailability	Low (2)	High (5)	(10) Medium	Aayam Regmi	Test hardware before testing, if hardware gives out during testing purchase backup hardware, borrow backup piano for project duration in case of failure	Closed
8	Time overrun due to integration & debugging complexity	Not being able to meet MS projects deadlines due to implementation and data collection	Medium (3)	High (5)	(15) High	Aayam Regmi	Follow RAD methodology and complete testing system before MS project deadline, Prioritize more time for data collection	Closed
9	Data loss or corruption during long recording sessions	Video datasets lost due to software crashes or memory corruption	Low (1)	Medium (3)	(3) Medium	Aayam Regmi	Have backup data in GitHub, rerecord data incase backup does not exist	Closed

Summary:

Likelihood:

- 4 low
- 5 medium

Impact:

- 1 low
- 4 medium
- 4 high

Severity:

- 1 low
- 6 medium
- 2 high